



Results

BIOPSY OR SURGICAL SPECIMEN (Order

Patient Information

Patient Name	MRN	Sex Male	DOB
--------------	-----	-------------	-----

Result Information

Status Edited	Provider Status Reviewed
------------------	-----------------------------

Entry Date

Component Results

Component <u>Surgical Pathology</u>	Lab
--	-----

Specimen

Procedure/Addenda
 Image Analysis Date Ordered:
 Date Complete:
 Date Reported:

Status: Signed Out

ICD-O-3
Carcinoma, urothelial NOS
8/20/13
Site Lateral wall of bladder
8/13/13
C67.2

Interpretation
 Prognostic markers performed on block M27

BLADDER CANCER RECURRENCE SCORE

p21 by IHC and IMAGE QUANTITATION:

Unaltered
 Percent Positive Average: 69%
 Staining Intensity Grade: 2
 HistoScore: 7
 p21 altered range (<10%)

p27 by IHC and IMAGE QUANTITATION:

Altered
 Percent Positive Average: 4%
 Staining Intensity Grade: 1
 HistoScore: 3
 p27 altered range (<30%)

p53 by IHC and IMAGE QUANTITATION:

Altered
 Percent Positive Average: 74%
 Staining Intensity Grade: 2
 HistoScore: 7
 p53 altered range (>=10%)

Ki-67 by IHC and Image QUANTITATION:

Altered

Percent Positive Average: 35%
Staining Intensity Grade: 2
HistoScore: 5
Ki-67 altered range (>10%)

Cyclin E by IHC and IMAGE QUANTITATION:

Unaltered
Percent Positive Average: 80%
Staining Intensity Grade: 2
HistoScore: 7
Cyclin E altered range (<30%)

INTERPRETATION:

Prognostic Significance: Unfavorable
3 Biomarker(s) altered
Range: Favorable <= 2 altered biomarkers
Unfavorable > 2 altered biomarkers

Comment:

In the above tumor profile the number of altered biomarkers improves the predictive accuracy for patients with PTa-T3 N0 Bladder Cancer. Greater than 2 altered biomarkers are associated with increased risk of recurrence.

Note:

Staining intensity based on 0 (neg) to 3+ (highest)
Percent Positive Average: Percentage of tumor cells staining.
HistoScore = Percentage of tumor cells staining (Scale 0-5) + Intensity (Scale 0-3).

These tests were performed IHC in conjunction with automated image analysis. The performance characteristics of the above tests have been determined in the While some antibodies have not been approved by the FDA, clearance / approval is not mandated. These antibodies are well documented and clinically accepted prognostic indicators.

These tests should not be regarded as part of research investigations. Known positive and negative control tissue show appropriate staining.

The above tumor bio-markers were performed per reflex protocol.

The system used for image quantitation is the

The method used for antigen retrieval is heat induced epitope retrieval.

College of American Pathologists (CAP) required information for predictive/prognostic markers. Type of specimen fixation and detection system: polymer detection kits are used on formalin-fixed paraffin-embedded sections.

Clones user: p21 (SX118); p27(SX53G8); p53 (DO-7); Ki-67 (MIB-1); Cyclin E (HE12)

Prognostic markers performed at the

Results-Comments
{Not Entered}

Final Diagnosis

- A. Right ureter, ureterectomy:
 - Reactive urothelium
 - No dysplasia or carcinoma identified
 - Frozen section diagnosis confirmed
- B. Left ureter, ureterectomy:
 - Benign ureter
 - No dysplasia or carcinoma identified
 - Frozen section diagnosis confirmed
- C. Left vas deferens, resection:
 - Fragment of vas deference with mostly denuded epithelium
- D. Left internal and external iliac lymph nodes, dissection:
 - Three lymph nodes, negative for tumor (0/3)
- E. Left obturator lymph nodes, dissection:
 - Three lymph nodes, negative for tumor (0/3)
- F. Left common iliac lymph nodes, dissection:
 - Five lymph nodes, negative for tumor (0/5)
- G. Preaortic lymph nodes, dissection:
 - Three lymph nodes, negative for tumor (0/3)
- H. Right common iliac lymph nodes, dissection:

- One lymph node, negative for tumor (0/1)

I. Precaval and paracaval lymph nodes, dissection:

- Six lymph nodes, negative for tumor (0/6)

J. Presacral lymph nodes, dissection:

- Two lymph nodes, negative for tumor (0/2)

K. Right internal and external lymph nodes, dissection:

- Two lymph nodes, negative for tumor (0/2)

L. Right obturator lymph nodes, dissection:

- Two lymph nodes, negative for tumor (0/2)

M. Bladder and prostate, radical cystectomy and prostatectomy:

- Invasive, high-grade papillary urothelial carcinoma extending into prostate parenchyma

- Prostatic adenocarcinoma, Gleason score 7 (4+3) (See attached CAP template)

- High-grade prostatic intraepithelial neoplasia

- Resection margins negative for tumor

Specimen: Bladder

Other (specify): Prostate, vas deferens, lymph node dissection, appendix

Procedure: Radical cystoprostatectomy

Tumor Site: Right lateral wall

Prostatic urethra and prostate

Tumor Size: Greatest dimension: 2.1 cm

Histologic Type: Urothelial (transitional cell) carcinoma

Associated Epithelial Lesions: None identified

Urothelial Carcinoma (WHO 2004/ISUP): High-grade

Microscopic Tumor Extension: Prostatic stroma

Margins: Margins uninvolved by invasive carcinoma

Lymph-Vascular Invasion: Not identified

Pathologic Staging (pTNM):

Primary Tumor (pT): pT4a: Tumor invades prostatic stroma

Regional Lymph Nodes (pN): pN0: No lymph node metastasis

Specify: Number examined:

27

Number involved (any size): 0

Distant Metastasis (pM): Not applicable

Additional Pathologic Findings: Adenocarcinoma of prostate (see attached separate CAP template)

Ancillary Studies: Block M27 sent for ancillary studies, report to follow.

Procedure: Other (specify): Cystoprostatectomy

Prostate Size: Weight: Not available
Size: 3.8 x 3.1 x 3.0 cm

Lymph Node Sampling: Pelvic lymph node dissection

Histologic Type: Adenocarcinoma (acinar, not otherwise specified)
Other (specify): Urothelial carcinoma (see above)

Histologic Grade

Gleason Pattern

Primary Pattern: Grade 3

Secondary Pattern: Grade 4

Tertiary Pattern: Not applicable
Total Gleason Score: 7

Tumor Quantitation
Proportion (percentage) of prostate involved by tumor: 5%

Extraprostatic Extension: Not identified

Seminal Vesicle Invasion: Not identified

Margins: Margins uninvolved by invasive carcinoma

Treatment Effect on Carcinoma: Not identified

Lymph-Vascular Invasion: Not identified

Perineural Invasion: Not identified

Pathologic Staging (pTNM)

Primary Tumor (pT): pT2a: Unilateral, involving one-half of 1 side or less

Regional Lymph Nodes (pN): pN0: No regional lymph node metastasis
Specify: Number examined: 27
Specify: Number involved: 0

Distant Metastasis (pM): Not applicable

Additional Pathologic Findings: High-grade prostatic intraepithelial neoplasia (PIN)

Ancillary Studies: Block M20 can be used for further studies if needed.

The above pathology diagnosis incorporates data elements as outlined in the CAP Cancer Protocols and Checklists which are based on the AJCC/UICC TNM, 7th edition.

N. Appendix, appendectomy:

- Obliterated appendix
- No malignancy identified

O. Ureter, left distal margin, ureterectomy:

- Benign fragment of ureter
- No dysplasia or carcinoma identified

P. Ureter, right distal margin, ureterectomy:
 - Benign fragment of ureter
 - No dysplasia or carcinoma identified

Note

This case was reviewed at the

Clinical History

Bladder cancer

Radical cystectomy, prostatectomy, lymph node dissection ileal conduit. All other indicated procedures

Source:

- A: Right ureter
- B: Left ureter
- C: Left vas deferens
- D: Left internal and external iliac lymph nodes
- E: Left obturator lymph nodes
- F: Left common iliac lymph nodes
- G: Preaortic lymph nodes
- H: Right common iliac lymph nodes
- I: Precaval and paracaval lymph nodes
- J: Presacral lymph nodes
- K: Right internal and external lymph nodes
- L: Right obturator lymph nodes
- M: Bladder and prostate
- N: Appendix
- O: Left distal margin (ureter)
- P: Right distal margin (ureter)

Gross Description

Specimen A, received fresh labeled with the patient's name and designated "right ureter", is an oriented tan tubular segment of tissue, 1.4 cm in length and 0.4 cm in diameter. There is a black silk suture at one end designated as distal per the surgeon. The opposing proximal end is submitted for frozen section as A1.

Specimen B, received fresh labeled with the patient's name and designated "left ureter", is an oriented tubular segment of tissue, 1.5 cm in length and 0.4 in diameter. There is a Prolene suture at one end designated as distal per the surgeon. The opposing proximal end is submitted for frozen section diagnosis as B1.

Specimen C, received in formalin labeled with the patient's name and designated "left vas deferens", is a portion of adipose tissue, 3.6 x 2.0 x 1.1 cm. One surface is pink and glistening and serosalized. The opposing surface is focally hemorrhagic adipose tissue. Specimen is serially sectioned showing focally hemorrhagic adipose tissue and a tubular vessel, consistent with a blood vessel. A discrete vas deferens is not identified. There are two cramped edges approximately 1.0 cm in length. The specimen is submitted entirely with the exception of the cramped edges at C1-C5.

Specimen D, received in formalin labeled with the patient's name and designated "left internal and external iliac lymph nodes", is fragmented fibrofatty tissue, 7.5 x 6.0 x 2.0 cm in aggregate. There are three possible nodes and a tan

7
tubular segment of tissue. The nodes are 0.5 cm to 2.0 x 0.7 x 0.4 cm. The tubular segment of tissue is 4.0 cm in length and 0.6 cm in diameter. The lumen is pinpoint and ill-defined. The wall surrounding the lumen is white and fibrous, not resembling lymph node parenchyma. Representative sections are submitted as follows:

Section key:

- D1 two whole nodes
- D2 one node serially sectioned into five pieces
- D3 tan tubular tissue submitted entirely
- D4 fibrous tissue for microscopic nodes

Specimen E, received in formalin labeled with the patient's name and designated "left obturator lymph nodes", is fragmented adipose tissue, 5.5 x 4.0 x 1.2 cm. There are three possible lymph nodes, 0.5 cm to 3.2 x 1.0 x 0.7 cm. The nodes are submitted entirely as follows:

Section key:

- E1 two whole nodes
- E2-E3 one node serially sectioned into eight pieces

Specimen F, received in formalin labeled with the patient's name and designated "left common iliac lymph nodes", is 4.0 x 2.5 x 1.0 cm of fragmented adipose tissue. There are three possible nodes, 0.6 cm to 2.0 cm. The nodes are submitted entirely as follows:

Section key:

- F1 two whole nodes
- F2 one node serially sectioned into five pieces
- F3 fibrous tissue for microscopic node

Specimen G, received in formalin labeled with the patient's name and designated "preaortic lymph nodes", is fragmented fibrofatty tissue with two staples. There is a 0.2 cm tan node adjacent to one of the staples. The specimen is submitted entirely as follows:

Section key:

- G1 one lymph node
- G2 fibrous tissue for microscopic nodes

Specimen H, received in formalin labeled with the patient's name and designated "right common iliac lymph nodes" is fragmented adipose tissue, 4.0 x 2.5 x 0.6 cm. There is one 0.3 cm tan node. Representative sections are submitted as follows:

- H1 - one node
- H2-H3 - fibrous tissue for microscopic nodes

Specimen I, received in formalin labeled with the patient's name and designated "precaval and paracaval lymph nodes" is fragmented adipose tissue, 4.0 x 3.5 x 1.2 cm. There are seven possible nodes, 0.3 cm to 1.2 cm. The specimen is submitted entirely as follows:

- I1 - five whole nodes
- I2 - two whole nodes

Specimen J, received in formalin labeled with the patient's name and designated "presacral lymph nodes" is fragmented adipose tissue, 3.2 x 2.4 x 1.1 cm. There are three possible lymph nodes, 0.2 cm to 1.2 cm. The nodes and representative fibrous tissue are submitted as follows:

- J1 - two whole nodes
- J2 - one node bisected
- J3 - fibrous tissue for microscopic nodes

Specimen K, received in formalin labeled with the patient's name and designated "right internal and external iliac lymph nodes" is fragmented adipose tissue, 3.0 x 3.0 x 0.8 cm. There are two lymph nodes, 0.3 cm and 1.9 cm. The nodes

are submitted entirely as follows:

K1 - one whole node

K2 - one node serially sectioned into five pieces

Specimen L, received in formalin labeled with the patient's name and designated "right obturator lymph nodes" is fragmented adipose tissue, 4.5 x 3.0 x 1.1 cm.

There are three possible nodes, 0.2 cm, 0.5 cm, and 5.0 x 1.2 x 1.0 cm. The nodes are submitted entirely as follows:

L1 - two whole nodes

L2-L3 - one node serially sectioned into nine pieces

Specimen M:

Type of specimen: Urinary bladder, ureter stumps, prostate and seminal vesicles

The patient's name and case number on the specimen container match the accompanying paperwork and cassettes.

Procedure: Radical cystectomy and prostatectomy

Fixative: Formalin

Dimensions:

Bladder - 6.0 cm - superior to inferior x 7.5 cm - medial to lateral x 3.5 cm - anterior to posterior

Right ureter - 1.6 cm in length, 0.4 cm in diameter

Left ureter - 1.3 cm in length, 0.4 cm in diameter

Prostate, 3.1 cm - apex to base, 3.8 cm - right to left x 3.0 cm - anterior to posterior

Description of findings:

Tumor: There is a tan, fluffy, lobulated tumor in the right lateral wall, 2.1 x 1.1 cm. The tumor extends through the mucosa into the muscle and is greater than 1.0 cm from the inked posterior adipose tissue.

Involvement of important adjacencies: None

Uninvolved mucosa: The mucosa in the left bladder is edematous.

Prostate, seminal vesicles and vasa deferentia: The prostate is serially sectioned apex to base into six slabs. The seminal vesicles are present. In slabs 3, 4 and 5, there is a firm nodule, 1.6 x 1.5 x 1.0 cm. This nodule has a well-defined smooth border. It does not appear to cross the midline into the right lobe.

Ink code:

Bladder anterior - blue, bladder posterior - orange, right prostate - orange, left prostate - blue

Section key:

M1 - right ureteral orifice to ureter edge

M2 - left ureteral orifice to ureter edge

M3, M4 - bladder neck, green ink at edge of tumor for orientation

M5-M12 - tumor submitted entirely from the right wall, superior to inferior

M13 - mucosa, right dome

M14 - mucosa, anterior wall

M15 - mucosa, left dome

M16, M17 - mucosa, left lateral wall

M18 - mucosa, posterior wall

M19 - slab 1, prostate, right apex, serially sectioned

M20 - slab 2, left

M21 - slab 4, right posterior

M22 - slab 5, right posterior

M23 - slab 6, right posterior

M24 - slab 6, right seminal vesicle, right seminal vesicle, prostatic interface

M25 - slab 1, left apex, serially sectioned

M26 - slab 2, left plus a trimming piece

M27 - slab 3, left

M28 - slab 4, left anterior

M29 - slab 4, left posterior

M30 - slab 5, left anterior

M31 - slab 5, left posterior

M32 - slab 6, left posterior with seminal vesicle prostatic interface (

Specimen N, received in formalin labeled with the patient's name and designated "appendix" is a vermiform appendix with attached mesoappendix. The appendix is 7.0 cm in length and 0.6 cm in diameter. The mesoappendix is 6.5 x 2.5 x 1.2 cm. The serosa is tan, smooth, and glistening. The cut surface of the appendix shows a 0.1 cm thick wall. the surface shows diffuse infiltrating adipose tissue extending from approximately 1.0 cm from the proximal margin to the distal tip. Representative sections are submitted as follows:
N1 cross sections, proximal margin inked orange distal tip
N2 representative cross sections

Specimen O, received in formalin labeled with the patient's name and designated "left distal ureter margin" is a strip of mucosa, 1.5 cm in length and 0.7 cm in diameter. There is a Prolene suture at one end of the strip of mucosa. This end of the specimen is inked orange. The specimen is bisected and submitted entirely as O1.

Specimen P, received in formalin labeled with the patient's name and designated "right distal ureter" is a roughly square portion of mucosa, 0.7 x 0.7 cm. There is a black silk suture on one side. The sutured side is inked orange. The specimen is bisected and submitted entirely as P1.

Microscopic Description

The stain quality is acceptable.

A-P. The microscopic findings are reflected in the diagnosis rendered.

Intraoperative Diagnosis

Time in: Time out:
A1-FS. Right ureter: "Negative for tumor or high-grade dysplasia."
B1-FS. Left ureter: "Negative for tumor or high-grade dysplasia."

Reported to

***Electronically Signed Out By

Results

Pathology Report - Scan on

Pathology Report - Scan on 1

: Bladder Cancer Recurrence Score

: Intraoperative Consultation Report

Result History

BIOPSY OR SURGICAL SPECIMEN

: Order Result History Report.

Collection Information

Specimen Source Date and Time
Other

Accession #

Lab Information

Criteria	Yes	No
Diagnosis Discrepancy		
Primary Tumor Site Discrepancy		
HPAA Discrepancy		
Prior Malignancy History		
Dual/Synchronous Primary		
Case is (circle):		
Reviewer Initials		
Date Reviewed		